MECH 325 Machine Elements

Gear Trains

Tutorial 5 – Bevel Gear Stress Analysis

1. Adapted from **Shigley, Problem 15-1 and 15-2**.

An uncrowned straight-bevel pinion has 20 teeth, a diametrical pitch of 6 teeth per inch and a transmission accuracy of 6. Both the pinion and gear are made of through-hardened steel with a Brinell hardness of 300. The driven gear has 60 teeth. The gearset has a life goal of 10^9 revolutions of the pinion with a reliability of 0.999. The shaft angle is 90° and the pinion speed is 900rpm. The face width is 1.25 inches, and the normal pressure angle is 20°. The pinion is mounted outboard of the bearings (cantilevered) while the gear is straddle mounted. Based on the AGMA bending strength, what is the power rating of the gearset. Assume overload factor is 1.0 and safety factors $S_F = S_H = 1.0$.

Part 1: To be demonstrated by the TA.

a. Determine the AGMA factors for bending
b. Determine the allowable AGMA bending strength number for the through-hardened pinion and gear.
c. Determine the power rating for the bevel gear set based on bending

Part 2: To be done by the student

Based on the AGMA contact (wear) strength, what is the power rating of the gearset. Assume overload factor is 1.0 and Safety Factors $S_F = S_H = 1.0$.

d. Determine the AGMA factors for contact stress
e. Determine the allowable AGMA contact strength number for the through hardened pinion and gear
f. Determine the overall power factor for the gear set. Based on both bending and wear, what is the maximum power allowed in the gear set?